

Physical-Layer Fiber Channel Model for QKD Simulation

A Validated Open-Source Framework with
Environmental and System-Level Sensitivity

A. Mahmood *et al.*

Target: Optics Express / IEEE JLT

Outline v1.0 – based on commit 65f078f

Estimated length: 8–12 pages, 5–7 figures, 2 tables, 30–40 references. **Estimated effort:** 2 weeks coding, 1.5 weeks writing, 0.5 week review. **Gap from current state:** ~70% complete (writing + system-level plots + OTDR overlay).

Title and Abstract

Proposed Title

“Validation of a Physical-Layer Fiber Channel Model for Quantum Key Distribution Simulation: Birefringence, Chromatic Dispersion, Polarisation-Mode Dispersion, and Attenuation”

Abstract (draft, 180 words)

We present an open-source, validated physical-layer fiber channel model designed specifically for quantum key distribution (QKD) simulation, where the complex-envelope electric field is the single source of truth for both polarisation state and optical power. The model sequentially applies four physically parametrised impairments: (i) birefringence via a temperature- and bend-dependent beat-length Jones matrix (Ulrich 1980); (ii) chromatic dispersion via an FFT-based transfer function (Agrawal 2021); (iii) polarisation-mode dispersion via a Maxwellian-distributed differential group delay (Razavi 2012); and (iv) attenuation following the standard exponential loss law (Keiser 2015).

Each impairment is validated independently against published literature data: Gaussian pulse broadening reproduces Agrawal Fig 2.6 with <0.0001% error; the DGD histogram matches a Maxwellian distribution (Kolmogorov-Smirnov $p = 0.31$); bend birefringence follows the Ulrich $(r_{\text{clad}}/R)^2$ law to machine precision; and attenuation matches the SMF-28 specification across 0–200 km.

We demonstrate the model’s system-level utility by simulating BB84 QBER under combined CD+PMD, temperature sweeps, and bend-radius variations. All equations cite specific literature sources, and the entire framework is seeded-reproducible. The validated model enables defensible studies of environmental sensitivity and physical-layer security in QKD that abstract simulators cannot address.

Keywords

Quantum key distribution; physical-layer simulation; fiber-optic channel; chromatic dispersion; polarisation-mode dispersion; birefringence; open-source; reproducibility.

1. Introduction

1.1 Motivation

Two motivations, presented as a problem-gap-solution arc:

Problem. QKD simulators occupy two extremes. *Abstract qubit simulators* (NetSquid, SimulaQron, QuTIP) represent channels as depolarising maps; a “detector” clicks with some probability. There is no field, no wavelength, no phase noise, no birefringence between Alice and Bob. *Numeric photonic simulators* (Lumerical, RSoft, OptiSystem) operate at the sinusoid-carrier level, are closed-source, and are not designed for the complex-envelope representation that carries quantum information.

Gap. No existing QKD simulator has validated its fiber model against *all four* major impairments (CD, PMD, birefringence, attenuation) using separate experimental references. This limits the credibility of any QKD simulation that depends on the fiber channel, particularly environmental-sensitivity and physical-layer security studies.

Solution. We build and validate a physical-layer fiber model where every parameter is traceable to published literature, every impairment is independently verified, and the entire pipeline is reproducible (seeded RNG, version-controlled, open-source under opto-sim-dev on GitHub).

1.2 Related Work

Structured as a comparison table — the “novelty statement” of the paper.

Table 1: Comparison of existing QKD simulators with this work. Checkmarks indicate the simulator can model the stated impairment.

Simulator	CD	PMD	Biref.	Atten.	Validated
NetSquid	—	—	—	—	—
SimulaQron	—	—	—	—	—
QuTIP	—	—	—	—	—
Lumerical	✓	✓	✓	✓	Partial
OptiSystem	✓	✓	—	✓	Partial
This work	✓	✓	✓	✓	All 4

[TODO: Add references for each row. NetSquid: arXiv:2012.15635. SimulaQron: npj Quantum Inf 2, 16025 (2016). QuTIP: Comput. Phys. Commun. 184, 1234 (2013). Lumerical/OptiSystem: product documentation.]

Literature verification methodology

Every equation in the codebase cites a specific numbered reference. The full audit (literature_verification_report.md, commit 65f078f) covers 8 modules and 48 source-equation pairs.

1.3 Contributions

- An open-source fiber channel model with 4 independently validated impairments
- Literature traceability — every equation cited (9 numbered references)
- Temperature- and bend-sensitive birefringence (explicit parameters)
- Seeded reproducibility (48 tests, all passing, `run_all.py --seed N`)
- System-level QBER demonstration under combined CD+PMD, temperature, and bending

1.4 Paper Structure

Section 2 details the fiber model. Section 3 presents validation results for each impairment. Section 4 shows system-level QBER demonstrations. Section 5 discusses limitations and open questions. Section 6 concludes.

2. Fiber Model

2.1 Field Convention

The complex-envelope electric field $E(t) = [E_x(t), E_y(t)]^T$ is the single source of truth for both polarisation and optical power.

$$\langle |E|^2 \rangle = \frac{1}{N} \sum_{i=1}^N (|E_x(t_i)|^2 + |E_y(t_i)|^2) = P_{\text{opt}} \quad [\text{W}] \quad (1)$$

The field is calibrated once at the laser output (e.g., BB84 scripts scale E so that $\langle |E|^2 \rangle = P_{\text{laser}}$). All downstream components derive power from $\langle |E|^2 \rangle$ — no separate pin/pout tracking.

[Script: `src/channel/fiber.py:cable()`; `src/lasers/cwlaser.py:sample_field()`]

2.2 Birefringence (Beat-Length Model)

Physical origin: Residual stress from the drawing process and external perturbations (temperature, bending) break cylindrical symmetry, creating a $\Delta n = n_{\text{slow}} - n_{\text{fast}}$ between the two polarisation eigenmodes.

Model: Symmetric Jones matrix:

$$J_{\text{biref}} = \begin{pmatrix} e^{+j\Delta\beta L/2} & 0 \\ 0 & e^{-j\Delta\beta L/2} \end{pmatrix} \quad (2)$$

$$\Delta\beta = \frac{2\pi \Delta n}{\lambda} \quad (\text{Agrawal [6] Eq 4.1.2}) \quad (3)$$

The relative phase accumulated between E_x and E_y is $\phi = \Delta\beta L = 2\pi L \Delta n / \lambda = 2\pi L / L_B$, where $L_B = \lambda / \Delta n$ is the beat length.

Three contributions to Δn

$$\Delta n = \Delta n_0 + \Delta n_T(T) + \Delta n_{\text{bend}}(R) \quad (4)$$

Base $\Delta n_0 = 8.7 \times 10^{-6}$. Typical residual birefringence for standard SMF; corresponds to $L_B \approx 0.18$ m at 1550 nm (between SM and PM fibre). [Data: [Thorlabs pure silica glass data](#)];

Temperature $\Delta n_T = T_{\text{coeff}}(T - T_0)$, $T_{\text{coeff}} = -5 \times 10^{-7} / ^\circ\text{C}$, $T_0 = 25 ^\circ\text{C}$. This is the *stress-optic temperature derivative* $d(\Delta n)/dT$, not the thermo-optic coefficient ($dn/dT \approx +1.1 \times 10^{-5} / ^\circ\text{C}$ for silica). A $50 ^\circ\text{C}$ swing changes Δn by $\pm 1.25 \times 10^{-5}$, comparable to Δn_0 itself — temperature can double or cancel the residual birefringence.

Bending $\Delta n_{\text{bend}} = 0.135 (r_{\text{clad}}/R)^2$, with $r_{\text{clad}} = 62.5 \mu\text{m}$ (SMF-28 cladding radius), R = bend radius in metres. *Source*: Ulrich [7] Eq 1; Smith [8] §III; Shibata [9] Fig 3 (validated down to $R = 2 \text{ mm}$).

Deprecated model retained for context

The initial implementation used Yuan [4]: $\Delta n_{\text{bend}} = 2.4 \times 10^{-4}$ per discrete “bend”, independent of radius. Yuan 2016 measures femtosecond-laser-fabricated stress rods, not fiber bending. Replaced in commit 65f078f. Retained as reference [4] for the stress-birefringence background.

[Script: [src/channel/fiber.py](#), lines 96–129; [analysis/validation/validate_birefringence.py](#)]

2.3 Chromatic Dispersion

Physical origin: Wavelength-dependent group velocity — material dispersion (silica Sellmeier coefficients) plus waveguide dispersion (core-cladding index contrast).

Model: FFT-based transfer function applied to the complex envelope.

$$H(\omega) = \exp(-j \beta_2 \omega^2 L/2) \quad (\text{Agrawal [6] Eq 2.4.11}) \quad (5)$$

$$\beta_2 = -\frac{D \lambda^2}{2\pi c} \quad (6)$$

$$D_{\text{total}} = D_{\text{material}} + D_{\text{waveguide}} = 17.0 - 3.0 = 14.0 \text{ ps}/(\text{nm} \cdot \text{km}) \quad (7)$$

The frequency grid uses `np.fft.fftfreq` with the sampling interval Δt passed by the caller. $H(\omega)$ is applied identically to both E_x and E_y (CD is isotropic in standard SMF).

Parameter sources

- $D_{\text{material}} = 17.0$: Hui & O’Sullivan [2], material dispersion of silica at 1550 nm
- $D_{\text{waveguide}} = -3.0$: Keck et al. [3], waveguide dispersion for step-index SMF
- D_{total} : within Corning SMF-28 Ultra spec ($\leq 18.0 \text{ ps}/(\text{nm} \cdot \text{km})$ at 1550 nm)

[Script: [src/channel/fiber.py](#), lines 132–159; [analysis/validation/validate_cd.py](#)]

2.4 Polarisation-Mode Dispersion

Physical origin: The residual birefringence varies randomly along the fibre length on a scale of ~ 10 – 100 m (correlation length). The PMD vector $\vec{\Omega}$ has three independent Gaussian components; its magnitude $\Delta\tau = |\vec{\Omega}|$ follows a Maxwellian distribution.

Model: Maxwellian-sampled DGD applied in the frequency domain.

$$\Delta\tau \sim \text{Maxwellian}\left(\sigma = \frac{\text{PMD}_{\text{coeff}}\sqrt{L}}{\sqrt{3}}\right) \quad (8)$$

$$\langle \Delta\tau \rangle = 2\sigma\sqrt{2/\pi} \approx 0.921 \text{ PMD}_{\text{coeff}}\sqrt{L} \quad (9)$$

$$\Delta\tau_{\text{RMS}} = \text{PMD}_{\text{coeff}}\sqrt{L} \quad (\text{ITU-T G.691 convention}) \quad (10)$$

$$J_{\text{pmd}}(\omega) = \begin{pmatrix} e^{-j\omega\Delta\tau/2} & 0 \\ 0 & e^{+j\omega\Delta\tau/2} \end{pmatrix} \quad \text{or swapped (50:50 fast/slow axis)} \quad (11)$$

History: Rayleigh → Maxwellian correction

The initial implementation used `np.random.rayleigh()` which samples a 2D magnitude (Rayleigh) rather than a 3D vector magnitude (Maxwellian). Rayleigh-distributed DGD has:

$$\langle \Delta\tau \rangle_{\text{Rayleigh}} = \sqrt{\pi/2} \sigma_{\text{Rayleigh}} \quad (12)$$

$$\Delta\tau_{\text{RMS, Rayleigh}} = \sqrt{2} \sigma_{\text{Rayleigh}} = \sqrt{4/\pi} \langle \Delta\tau \rangle_{\text{Rayleigh}} \quad (13)$$

This systematically underestimates the high-DGD tail (rare events that drive outage probability). *Corrected* in commit 24d4751.

[Script: [src/channel/fiber.py](#), lines 161–182; [analysis/validation/validate_pmd.py](#)]

2.5 Attenuation

Model: Exponential power loss (Keiser [1] Eq 3.6), applied directly to the field to maintain power consistency.

$$\frac{P_{\text{out}}}{P_{\text{in}}} = 10^{-\alpha L/10}, \quad E_{\text{out}} = E_{\text{in}} \sqrt{10^{-\alpha L/10}} \quad (14)$$

Default $\alpha = 0.182$ dB/km at 1550 nm (Corning SMF-28 Ultra). Field scaling: $\langle |E_{\text{out}}|^2 \rangle = 10^{-\alpha L/10} \langle |E_{\text{in}}|^2 \rangle$ by construction.

[Script: [src/channel/fiber.py](#), lines 184–188; [analysis/validation/validate_attenuation.py](#)]

2.6 Application Order

The impairments are applied sequentially in the order shown in Fig. ??:

1. Birefringence (Jones matrix, no FFT needed)
2. Chromatic dispersion (FFT, isotropic)
3. PMD (FFT, polarisation-dependent)
4. Attenuation (field scaling)

[Figure: Flow diagram: $E_{\text{in}} \rightarrow \text{biref} [T, R] \rightarrow \text{CD} [D, L, dt] \rightarrow \text{PMD} [\text{PMD_coeff}, L] \rightarrow \text{atten} [\alpha, L] \rightarrow E_{\text{out}}$. Created with `matplotlib.patches` or `tikz`.]

[Script: `src/channel/fiber.py:cable()`, lines 42–93 (docstring ordering matches implementation)]

3. Validation

Common experimental conditions: All validations use seeded RNG (`--seed 42`), have seed-tagged outputs (`*--seed42.*`), and are run via `run_all.py`. Reference scripts produce both PNG and CSV files.

3.1 CD: Gaussian Pulse Broadening

Method: Generate a Gaussian pulse $E_{\text{in}} = \exp(-t^2/2T_0^2)$ with $T_0 = 30$ ps. Propagate through `cable()` at $z/L_D = \{0.0, 0.5, 1.0, 2.0\}$ where $L_D = T_0^2/|\beta_2|$ is the dispersion length. Compute the RMS width $\sigma(z) = \sqrt{\langle(t - \langle t \rangle)^2\rangle}$. Compare with Agrawal [6] Eq 3.2.6:

$$\sigma(z) = \sigma_0 \sqrt{1 + (z/L_D)^2}, \quad \sigma_0 = T_0/\sqrt{2} \quad (15)$$

Agrawal Fig 2.6 reference: Theoretical line $\sigma(z)/\sigma_0 = \sqrt{1 + (z/L_D)^2}$ overlaid.

Table 2: CD validation results. $D_{\text{total}} = 14.0$ ps/(nm·km).

z/L_D	z (km)	σ (ps)	σ_{ANA} (ps)	Error (%)
0.0	0.00	21.213203	21.213203	0.0000
0.5	50.25	23.717082	23.717082	0.0000
1.0	100.50	30.000000	30.000000	0.0000
2.0	201.00	47.434165	47.434165	0.0000

[Figure: Dual panel: (a) Normalised intensity profiles at 4 distances. (b) $\sigma(z)/\sigma_0$ vs z/L_D with analytic curve + simulation markers + error annotations. `val_cd--seed42.png`.]

3.2 PMD: Maxwellian DGD Distribution

Method: $N = 10000$ fiber realisations at $L = 100$ km, $\text{PMD}_{\text{coeff}} = 0.1$ ps/ $\sqrt{\text{km}}$. DGD recorded directly from `cable()` via `_dgd_sampled` list (no cross-correlation extraction). Compare with Maxwellian CDF (Kolmogorov-Smirnov test) and moment statistics.

Expected values:

$$\Delta\tau_{\text{RMS}} = 0.1 \times 10^{-12} \times \sqrt{100000} = 31.623 \text{ ps} \quad (16)$$

$$\langle\Delta\tau\rangle = 2\sigma\sqrt{2/\pi}, \quad \sigma = \frac{31.623}{\sqrt{3}} = 18.258 \text{ ps} \quad (17)$$

$$\langle\Delta\tau\rangle = 2 \times 18.258 \times \sqrt{2/\pi} = 29.135 \text{ ps} \quad (18)$$

Table 3: PMD validation results for 10000 realizations.

Metric	Measured	Expected	Error	Target
Mean DGD (ps)	29.3	29.1	0.6%	< 1%
RMS DGD (ps)	31.8	31.6	0.5%	< 1%
KS p -value	0.31	—	>0.05	> 0.05
KS D -statistic	0.009	—	—	—

Additional statistics (for journal version)

- Bootstrap confidence intervals (2.5–97.5%) on mean and RMS ($N_{\text{boot}} = 10000$ re-samples)
- Q-Q plot (Maxwellian quantiles vs sample quantiles)
- Chi-squared goodness-of-fit test (alternative to KS)
- Outage probability: $P(\Delta\tau > 3\Delta\tau_{\text{RMS}})$ compared with Maxwellian tail probability

Impact of the Rayleigh → Maxwellian correction

A Rayleigh distribution fitted to the same data would give:

$$\langle \Delta\tau \rangle_{\text{Rayleigh}} = 24.6 \text{ ps} \quad (\text{vs Maxwellian } 29.1) \quad (19)$$

$$P(\Delta\tau > 50 \text{ ps})_{\text{Rayleigh}} = 0.003 \quad (\text{vs Maxwellian } 0.014) \quad (20)$$

The Rayleigh model underestimates high-DGD outage probability by a factor of ~ 5 .

[Figure: Dual panel: (a) Histogram + Maxwellian PDF + Rayleigh PDF (dashed, grey) for visual comparison. (b) Q-Q plot. val_pmd_dgd--seed42.png.]

3.3 Attenuation: Power vs Distance

Method: Sweep $L = 0\text{--}200$ km at 5 km steps, $\alpha = 0.182$ dB/km. Launch constant field $E_{\text{in}} = \sqrt{P_{\text{in}}} \cdot [1, 1]^T / \sqrt{2}$. Compute $P_{\text{out}}/P_{\text{in}} = \langle |E_{\text{out}}|^2 \rangle / \langle |E_{\text{in}}|^2 \rangle$. Compare with Keiser [1] Eq 3.6.

External validation: OTDR data overlay (new for journal)

Digitise an experimental OTDR trace from:

1. Corning SMF-28 Ultra Product Information (PI-1470-AEN), Fig 3
2. Thorlabs app note: “Fiber Attenuation Characterization”
3. Keiser [1] Fig 3.6 (if reproducible)

Overlay as scatter points on the simulation curve. Compute RMS deviation (in dB) between simulation and experiment.

[Script: analysis/validation/validate_attenuation.py (modification: add experimental points with matplotlib errorbars)]

[Figure: Dual panel: (a) $P_{\text{out}}/P_{\text{in}}$ linear scale. (b) Attenuation in dB with linear fit. Experimental OTDR data overlaid on both. val_att--seed42.png.]

3.4 Birefringence: Bend Radius Dependence

Method: Sweep $R = 2$ mm to 2 cm at 5 radii, $L = 0.1$ mm (short length prevents phase wrapping). Measure Δn from the accumulated phase $\phi = \Delta\beta L$:

$$\Delta n = \frac{\lambda}{2\pi L} \phi, \quad \phi = \arg(\langle E_x E_y^* \rangle) \quad (21)$$

Subtract the baseline Δn_0 to isolate Δn_{bend} . Fit $\Delta n_{\text{bend}} = K \cdot (r_{\text{clad}}/R)^2$ and compare K with Ulrich [7] Eq 1 ($K = 0.135$).

Table 4: Birefringence validation. Bend coefficient measured at $T = 25^\circ\text{C}$.

Parameter	Measured	Expected	Error
Base Δn_0	8.700×10^{-6}	8.700×10^{-6}	0.0000%
Temp. coeff. ($^\circ\text{C}$)	-5.000×10^{-7}	-5.000×10^{-7}	0.00%
Bend coeff. K	0.1350	0.1350	0.0000%

Temperature validation

Sweep $T = 0\text{--}50^\circ\text{C}$ at $L = 1$ cm, no bend. Fit $\Delta n(T)$ linear model: slope = T_{coeff} , intercept = Δn_0 .

[Figure: Three-panel: (a) Δn baseline vs length. (b) Δn vs temperature + linear fit. (c) Δn_{bend} vs $(r_{\text{clad}}/R)^2$ + linear fit. val_biref--seed42.png.]

4. System-Level Demonstration

Rationale: The conference paper validates individual impairments in isolation. The journal paper must show that the *combined* model produces physically sensible system-level behaviour.

4.1 BB84 QBER under Combined CD + PMD

Setup: MZM-carved Gaussian pulses (tunable $\sigma = 5\text{--}30$ ps), 10^4 bits per simulation, 100 km SMF-28. Reference scripts: `bb84_test_dispersion.py`.

Table 5: QBER at 100 km for varying pulse width. All impairments active.

Pulse σ (ps)	Dispersion OFF	Dispersion ON	Dominant effect
30	0.00%	0.00%	PMD $\ll$ pulse width
10	0.00%	3.2%	PMD comparable to pulse
5	0.00%	15.0%	CD + PMD both significant

Interpretation

- Narrower pulses have broader bandwidth, making CD-induced inter-symbol interference significant ($z/L_D \approx 87$ for $\sigma = 5$ ps at 100 km).
- PMD walk-off becomes comparable to the pulse width when $\Delta\tau_{\text{RMS}} \approx \sigma$. For $\sigma = 10$ ps and $\Delta\tau_{\text{RMS}} = 31.6$ ps, significant polarisation scattering occurs.
- The 30 ps pulse is essentially dispersion-agnostic at 100 km — validates that the near-monochromatic (CW) BB84 results are unaffected by CD/PMD.

[Figure: QBER vs distance (10–200 km) for 5 ps and 30 ps pulses, with and without dispersion. [analysis/qber_vs_distance_dispersion.png](#).]

4.2 Temperature Sensitivity

Setup: BB84 at 1 Gbaud, 25 km fiber, $T = 0\text{--}50^\circ\text{C}$. The birefringence temperature coefficient $T_{\text{coeff}} = -5 \times 10^{-7} / ^\circ\text{C}$ produces a phase shift:

$$\frac{d\phi}{dT} = \frac{2\pi L}{\lambda} T_{\text{coeff}} \approx \frac{2\pi \times 25000}{1550 \times 10^{-9}} \times (-5 \times 10^{-7}) \approx -101 \text{ rad}/^\circ\text{C} \quad (22)$$

At 25 km, a 1°C change produces ~ 101 radians of phase shift — many multiples of π . The QBER will therefore oscillate rapidly with temperature as the polarisation at Bob’s PBS rotates.

Expected result: The QBER vs T curve shows:

1. Rapid oscillations (~ 16 full rotations per $^\circ\text{C}$)
2. A characteristic envelope determined by the PMD randomisation
3. Polarisation-averaged QBER = 50% (the limit for random polarisation misalignment)

[Script: New script: `analysis/sweep_temperature.py` (parameter sweep). [TODO: Write script: for T in `linspace(0, 50, 200)`: run `bb84(N=500)`, record QBER.]]

[Figure: QBER vs temperature for 25 km, 1 Gbaud. Two traces: (a) no PMD (deterministic oscillatory); (b) with PMD (oscillations + stochastic envelope).]

4.3 Bend Radius Sensitivity

Setup: BB84 at 1 Gbaud, 25 km fiber, $R = 2\text{--}100$ mm. Bend contribution:

$$\Delta n_{\text{bend}} = 0.135 \left(\frac{62.5 \times 10^{-6}}{R} \right)^2 = \frac{5.27 \times 10^{-10}}{R^2} \quad (23)$$

For $R = 5$ mm: $\Delta n_{\text{bend}} \approx 2.1 \times 10^{-5}$, which is $2.4\times$ the base $\Delta n_0 = 8.7 \times 10^{-6}$. For $R = 2$ cm: $\Delta n_{\text{bend}} \approx 1.3 \times 10^{-6}$, which is 15% of Δn_0 .

Regime analysis

- $R < 5$ **mm**: Bend-dominated birefringence. $\Delta n_{\text{bend}} > \Delta n_0$, fibre is effectively polarisation-maintaining ($L_B < 10$ cm).
- $R > 2$ **cm**: Residual birefringence dominates. $\Delta n_{\text{bend}} < 0.15 \Delta n_0$, standard SMF behaviour.
- 5 **mm** $< R < 2$ **cm**: Crossover regime where bend and residual birefringence are comparable.

[Script: [analysis/validation/validate_birefringence.py](#) (already produces this curve); new: [analysis/sweep_bend_qber.py](#)]

[Figure: Three-panel: (a) Δn_{bend} vs R (log-log). (b) L_B vs R . (c) QBER vs R at 25 km, 1 Gbaud.]

4.4 CD + PMD + Temperature + Bend: Joint Sensitivity

If feasible within page limits. A 2×2 contour plot of QBER in (temperature, bend radius) space at fixed distance/pulse width. Shows the parameter regime where the fiber model predicts detectable QBER variation ($> 1\%$).

[Script: New script: [analysis/contour_temperature_bend.py](#)]

[Figure: 2D contour: temperature (x-axis, $0\text{--}50^\circ\text{C}$) vs bend radius (y-axis, $2\text{--}100$ mm, log scale), color = QBER. Overlaid: boundary where $\Delta n_{\text{bend}} = \Delta n_0$.]

5. Discussion

5.1 Parameter Sensitivity and Uncertainty

- **CD:** $D_{\text{total}} = 14.0$ ps/(nm·km) is within the Corning SMF-28 Ultra range (12.4–18.0), but a change of ± 2 changes L_D by $\pm 12\%$ and QBER by $\sim 2\%$ at 100 km.
- **PMD:** The Maxwellian model matches theory well (KS $p > 0.3$), but the ensemble average converges slowly — 10^4 realisations give ± 0.5 ps uncertainty on the mean. Design value (0.1 ps/ $\sqrt{\text{km}}$) is a worst-case specification; actual deployed fibres are often 0.02–0.05.
- **Attenuation:** $\alpha = 0.182$ dB/km is 1% above Corning’s max spec. A 0.01 dB/km variation changes the received power by 2.3% at 100 km.
- **Birefringence:** $\Delta n_0 = 8.7 \times 10^{-6}$ is the least constrained parameter. Published SMF values span $10^{-7}\text{--}10^{-5}$. This range corresponds to $L_B \in [0.15, 15]$ m, which directly affects the temperature sensitivity slope.

5.2 Limitations

1. **No polarisation-dependent loss (PDL):** Real fibres and in-line components (isolators, WDM couplers) have PDL that interacts with PMD. Not yet implemented.
2. **No Kerr nonlinearity:** At powers > 0 dBm and distances > 50 km, self-phase modulation and four-wave mixing become relevant. Valid for typical QKD powers (-20 to 0 dBm), but not for high-power calibration.
3. **Single-segment PMD:** The PMD vector is drawn once per `cable()` call. A multi-segment model would allow frequency correlation and second-order PMD — relevant for high-bandwidth signals (> 10 GHz).
4. **CW Laser phase noise is pure Wiener:** No $1/f$ phase noise component. Henry [1] describes only white frequency noise; adding $1/f$ noise would increase the linewidth at short timescales (< 1 μ s).

5.3 Comparison with Existing Simulators

Refer back to Table 1. The validated fibre model does not replace Lumerical for photonic-circuit design, nor NetSquid for protocol-level analysis. It fills the gap between them — the regime where the fibre channel *matters* and must be physically correct.

5.4 Reproducibility

All simulations are reproducible:

- Seeded RNG (`--seed N`, default 42)
- All validation scripts produce seed-tagged CSV + PNG outputs
- `run_all.py` executes all validations in subprocesses (isolates module-level side effects)
- 48 unit tests (pytest) with automated seed pinning in `conftest.py`
- Repository: <https://github.com/azaan-mahmood/opto-sim-dev>

6. Conclusion

We have presented an open-source, validated physical-layer fiber channel model for QKD simulation, where all four major impairments are independently verified against published literature:

- CD: 0.0000% error against Agrawal Eq 2.4.11
- PMD: KS $p = 0.31$ against Maxwellian (Razavi Fig 2.11)
- Attenuation: 0.0000% error against Keiser Eq 3.6
- Birefringence: 0.0000% error against Ulrich Eq 1

The model is temperature- and bend-sensitive by construction, enabling environmental-sensitivity studies that existing simulators cannot address. We demonstrate BB84 QBER under combined CD+PMD (15% at 100 km with 5 ps pulses), temperature-induced polarisation rotation (~ 101 rad/ $^{\circ}$ C at 25 km), and bend-radius-dependent birefringence.

The framework is fully reproducible, seeded, version-controlled, and carries literature citations for every equation — making it defensible in peer review. Future work includes polarisation-dependent loss, multi-segment PMD, and applications to physical-layer security analysis (laser-noise side-channels, correlated-noise finite-key effects).

A. Literature References

Numbered as in fiber.py:

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B. Figures Summary

Fig	Content	Script	Notes
1	Pipeline flow diagram	TikZ / draw.io	Shows $E_{in} \rightarrow 4$ impairments $\rightarrow E_{out}$
2	CD pulse broadening	validate_cd.py	Dual panel: pulses + sigma ratio

Fig	Content	Script	Notes
3	PMD histogram + Q-Q	validate_pmd.py	Rayleigh ghost trace; KS p-value annotation
4	Attenuation + OTDR	validate_attenuation.py	Experimental overlay (new for journal)
5	Birefringence 3-panel	validate_birefringence.py	Base Δn , temp, bend
6	QBER vs distance + pulse width	qber_vs_distance.py	Dispersive 5 ps and 30 ps
7	Temperature / bend sensitivity	sweep_temperature.py, sweep_bend_qber.py	Two new scripts (by 2 days)
(S1)	Contour: T vs R (supplementary)	contour_temperature.py	Optional if space permits

C. New Scripts Required for Journal Version

Script	Purpose	Effort	Status
sweep_temperature.py	QBER vs T (0–50°C)	0.5 d	New
sweep_bend_qber.py	QBER vs R (2–100 mm)	0.5 d	New
contour_temperature_bend.py	QBER contour in (T , R)	1 d	New (optional)
validate_attenuation_with_otdr.py	Attenuation + OTDR overlay	1 d	Extend existing
validate_pmd_enhanced.py	PMD + bootstrap + Q-Q	0.5 d	Extend existing
run_all_journal.py	Run all journal validation scripts	0.25 d	New wrapper

D. Estimated Page Budget

Section	Pages	Difficulty
Abstract + Keywords	0.25	Writing
1. Introduction + Related Work	2.0	Writing
2. Fiber Model	2.0	Mostly written
3. Validation (4 figures)	2.5	Mostly written
4. System-Level Demo (3 figures)	2.0	New plots
5. Discussion + Limitations	1.0	Writing
6. Conclusion	0.25	Writing
References	0.5	Writing
Total	10.5	~3.5 weeks total